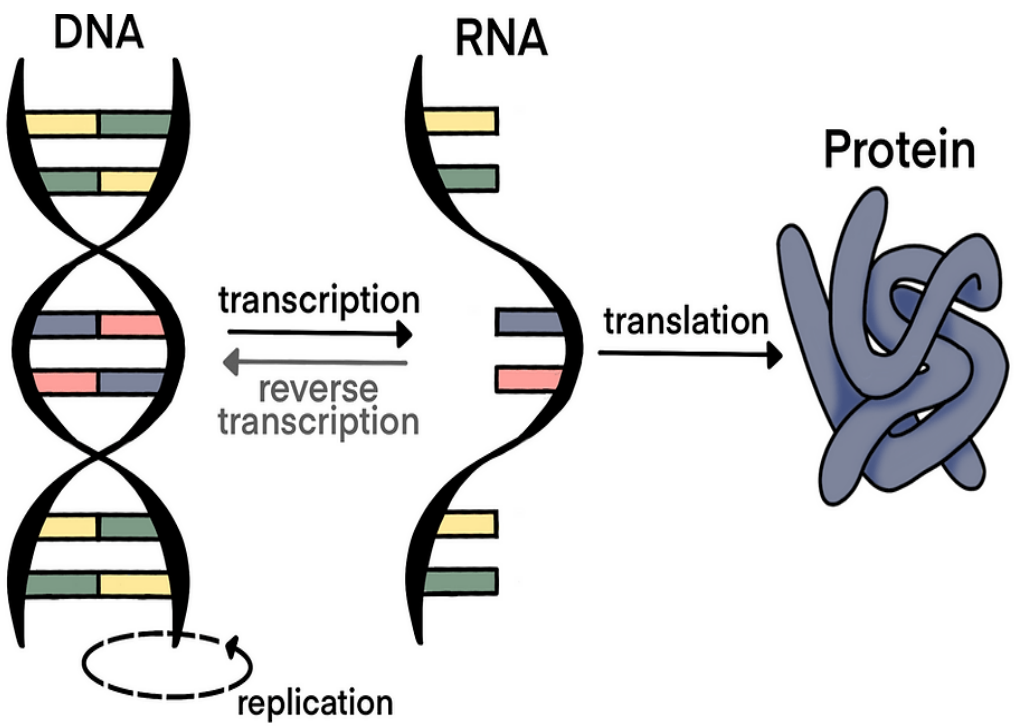
Module 6

Information Transfer

OVERVIEW OF GENETIC INFORMATION FLOW

Central Dogma of Molecular Biology
Proposed by Francis Crick, it explains how genetic information flows from DNA to RNA to Protein.

$$\text{DNA} \xrightarrow{\text{transcription}} \text{RNA} \xrightarrow{\text{translation}} \text{Protein}$$



KEY PROCESSES IN GENETIC INFORMATION FLOW

1.Replication:

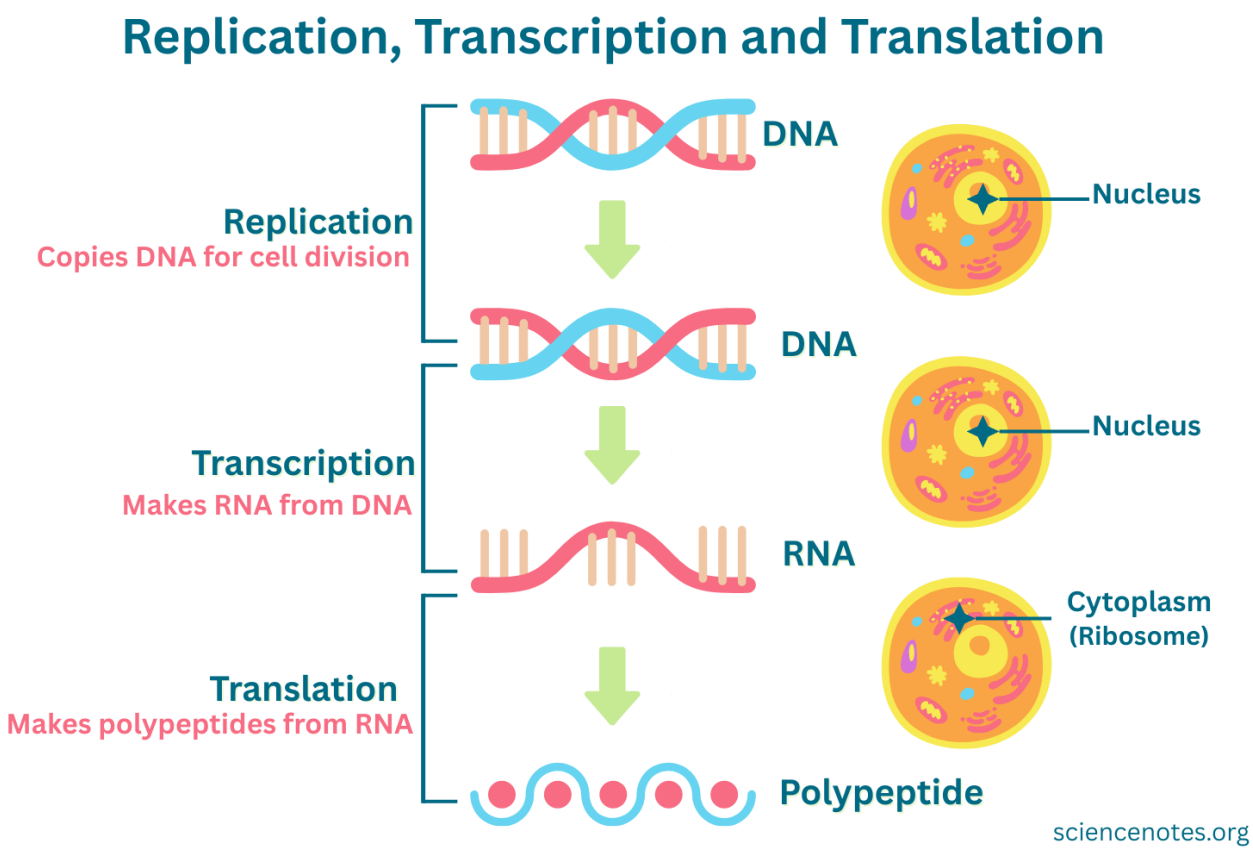
1. DNA copies itself during cell division.
2. Enzyme: **DNA polymerase**
3. Semi-conservative mechanism

2.Transcription:

1. DNA → RNA
2. Enzyme: **RNA polymerase**
3. Occurs in the nucleus (eukaryotes)
4. Produces: **mRNA (messenger RNA)**, **tRNA**, and **rRNA**

3.Translation:

1. mRNA → Protein
2. Occurs in the cytoplasm on **ribosomes**
3. Involves: **tRNA** carrying amino acids, **ribosomal RNA**, and **translation factors**



Frederick Griffith's Experiment (1928) – *Discovery of the Transforming Principle*

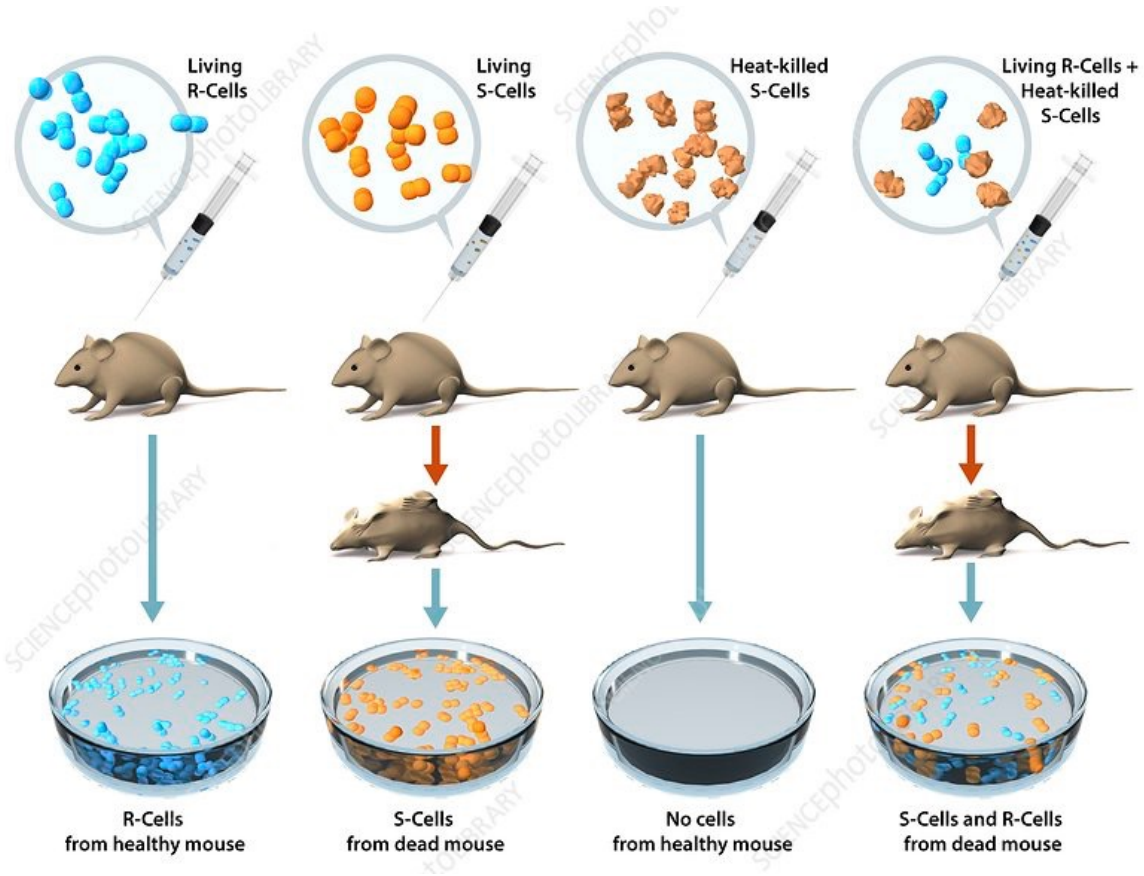
Objective: To understand how bacteria cause disease.

Organism used: *Streptococcus pneumoniae*, which has two strains:

- S strain (Smooth): Virulent (has a polysaccharide capsule, causes disease).
- R strain (Rough): Non-virulent (no capsule, does not cause disease).

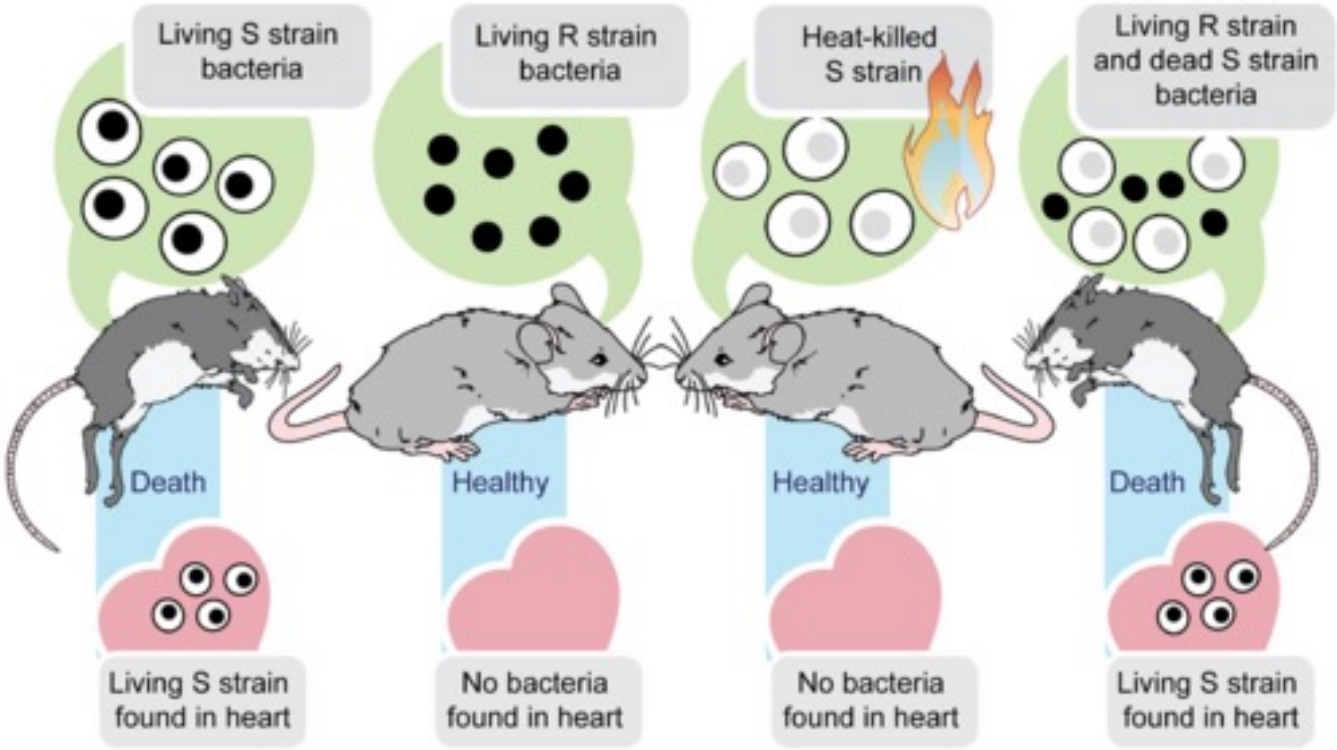
Experimental observations:

- Live S strain → **Mouse dies.**
- Live R strain → **Mouse survives.**
- Heat-killed S strain → **Mouse survives.**
- Live R strain + heat-killed S strain → **Mouse dies, and live S strain recovered.**



Conclusion (Griffith's Experiment):

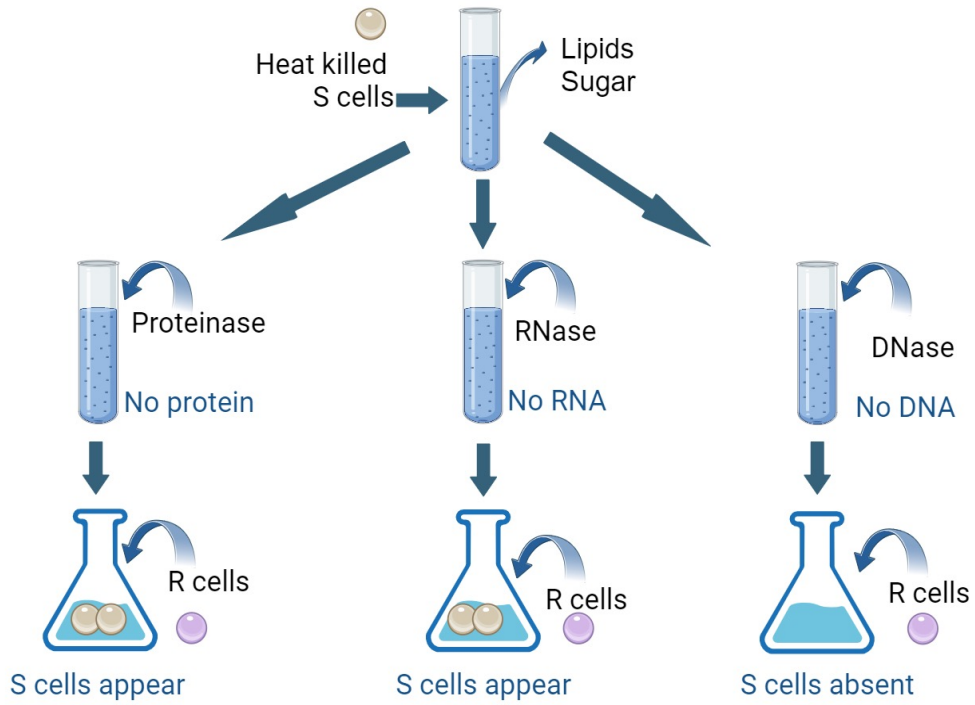
- A **substance** (later identified as DNA) from the **heat-killed virulent S strain** was **taken up by the live non-virulent R strain**.
- This **enabled the R strain to acquire the virulent trait** and transform into the **S-type**, capable of causing disease.
- The experiment **demonstrated that genetic traits can be transferred between bacteria** through a **heritable chemical substance**, laying the foundation for the discovery that **DNA is the genetic material**.



Avery, MacLeod, and McCarty (1944) – *Identification of DNA as the Transforming Substance*

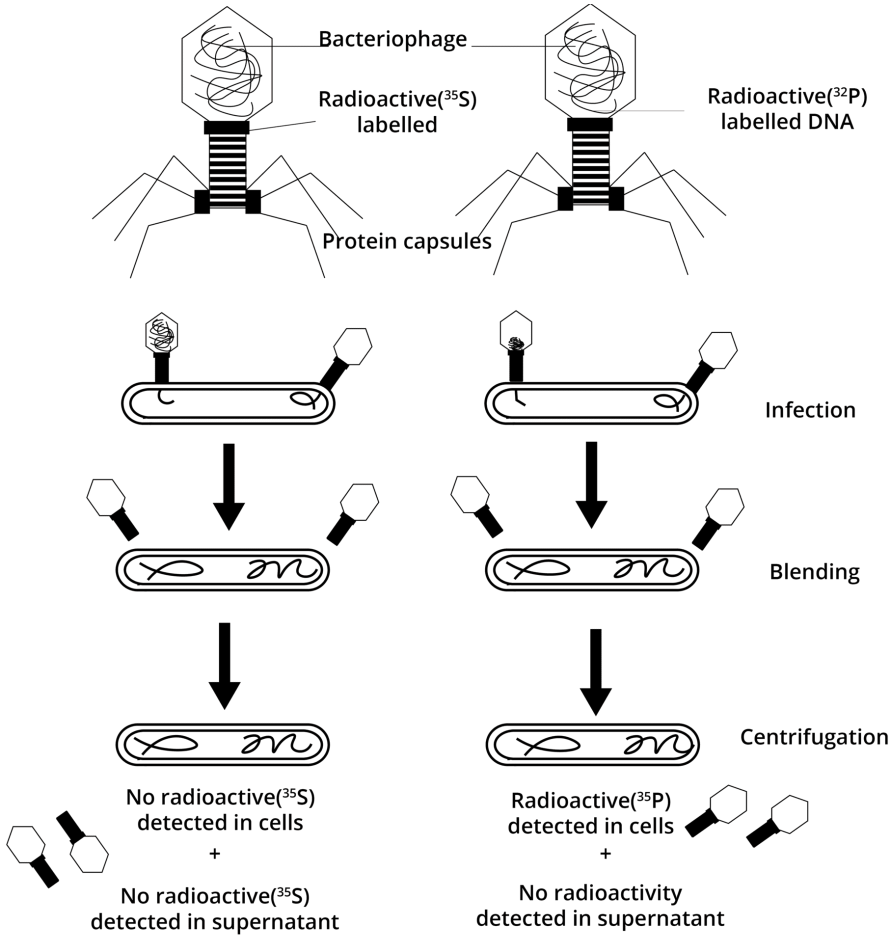
- **Objective**: To determine the chemical nature of Griffith's "transforming principle."
- **Approach**:
  - Used purified extract from heat-killed S strain.
  - Treated extracts with different enzymes to selectively destroy:
    - **Proteins** (using protease)
    - **RNA** (using RNase)
    - **DNA** (using DNase)
- **Results**:
  - **Transformation occurred** when treated with protease and RNase.
  - **Transformation failed** only when treated with **DNase**.
- **Conclusion**:
  - The **transforming principle is DNA**, not protein or RNA.
  - This was the **first direct experimental proof** that DNA is the genetic material.

Avery, MacLeod and McCarty's Experiment (1944)



Hershey and Chase Experiment (1952) – *Confirmation Using Bacteriophages*

- **Objective:** To confirm whether DNA or protein is the genetic material using viruses.
- **System used: Bacteriophage T2** (a virus that infects bacteria).
- **Strategy:**
  - Labeled viral DNA with radioactive phosphorus (^{32}P), as DNA contains phosphorus.
  - Labeled viral protein with radioactive sulfur (^{35}S), as proteins contain sulfur (not DNA).
- **Procedure:** Allowed labeled phages to infect E. coli. After infection, used a blender to separate viral coats from bacterial cells. Measured radioactivity in the cells and in the supernatant.
- **Results:**
  - ^{32}P was found inside the bacterial cells.
  - ^{35}S remained in the supernatant (outside the cells).
- **Conclusion:**
  - DNA entered the bacteria and directed viral replication.
  - Protein did not enter, confirming DNA is the genetic material.



HIERARCHY OF DNA STRUCTURE

Single-Stranded DNA (ssDNA):The Primary Level

Definition: Single-stranded DNA is a linear polymer composed of nucleotide monomers linked by phosphodiester bonds.

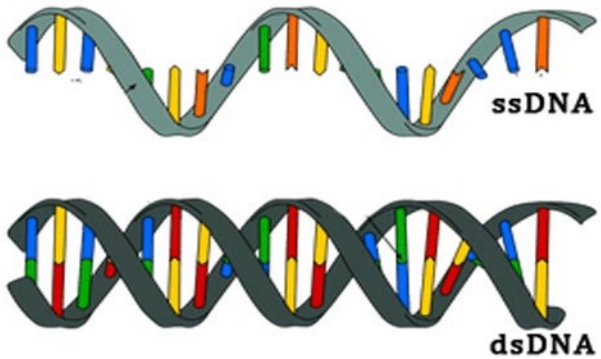
Components of a Nucleotide:

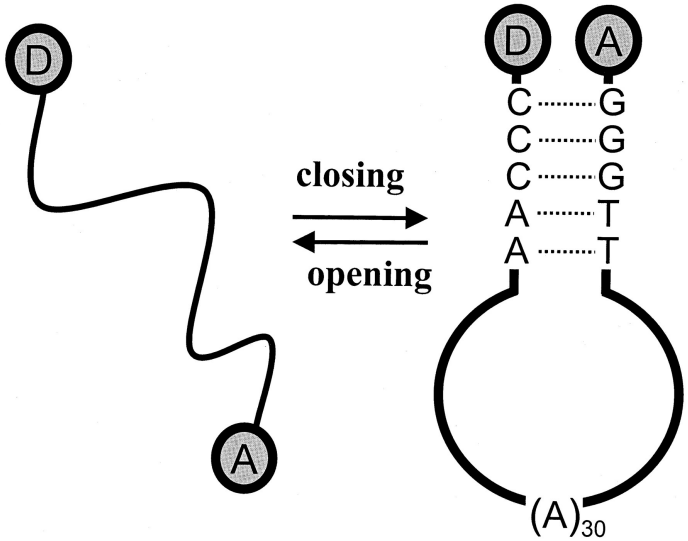
- Phosphate group
- Deoxyribose sugar
- Nitrogenous base (Adenine, Thymine, Cytosine, Guanine)

Physiological Relevance: ssDNA is transiently formed during processes such as DNA replication, transcription, and repair.

Flexibility: It lacks base-pairing, making it highly flexible and susceptible to nucleases.

Secondary Structures: Can form hairpins or stem-loop structures due to intrastrand base-pairing in certain sequences.





Hairpins or stem-loop structure

Double-Stranded DNA (dsDNA)

Formation: Two complementary ssDNA strands align in antiparallel orientation.

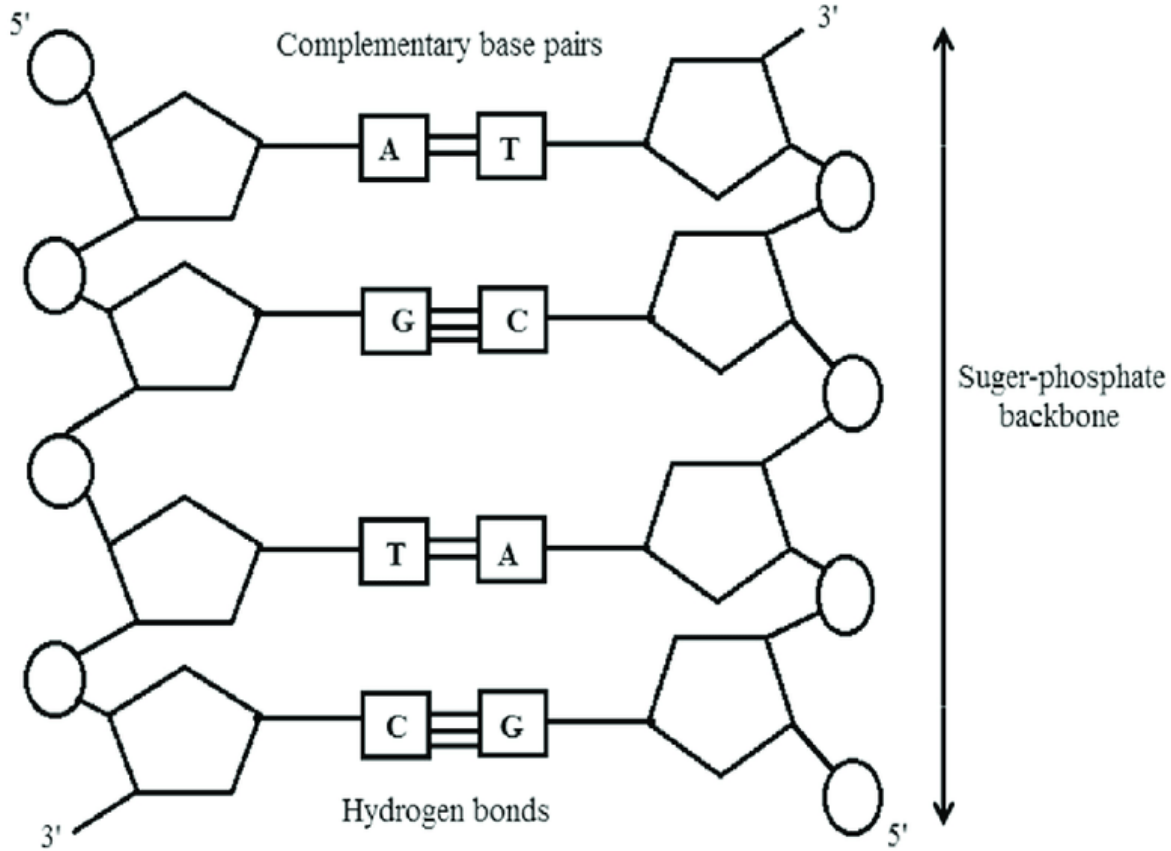
Base pairing via hydrogen bonds:

- A = T (2 H-bonds)
- G ≡ C (3 H-bonds)

Double Helix Characteristics:

- Distance between adjacent base pairs: ~0.34 nm
- 3.4 nm per helical turn
- Stabilized by hydrogen bonds and base stacking interactions.

Biological Role: Encodes genetic information and serves as a template for replication and transcription.



Nucleosome: Tertiary Packaging of DNA

Definition: The nucleosome is the basic structural unit of chromatin, consisting of DNA wrapped around a histone octamer.

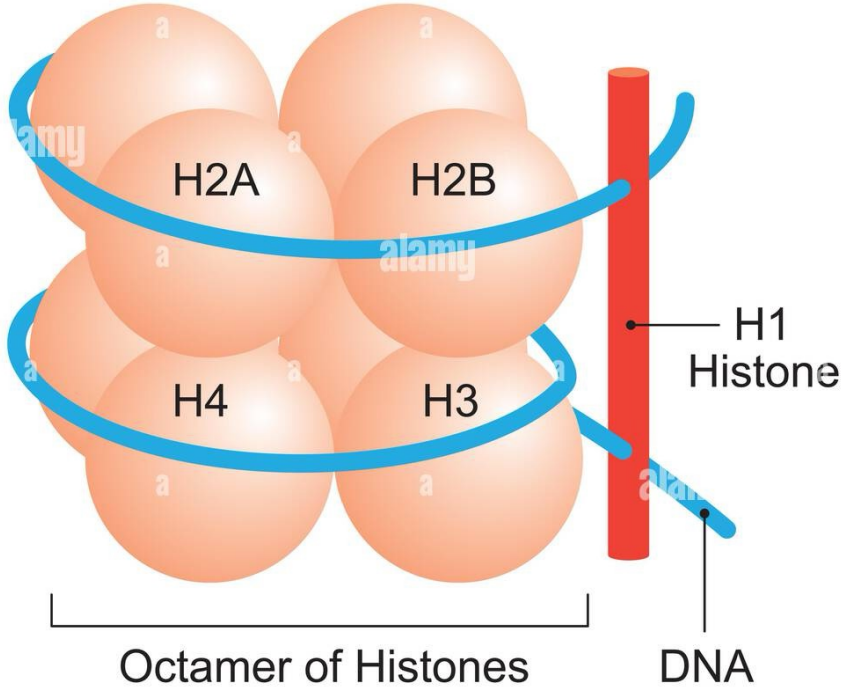
Composition:

- 147 base pairs of DNA
- Histone octamer: 2 copies each of H2A, H2B, H3, and H4
- H1 histone (linker histone) binds to the linker DNA region between two nucleosomes and to the nucleosome core at the DNA entry and exit sites. It stabilizes the nucleosome.

Functional Role:

- Compacts ~2 meters of genomic DNA into the nucleus (~10 µm diameter)
- Regulates gene accessibility and epigenetic modifications (e.g., acetylation, methylation)

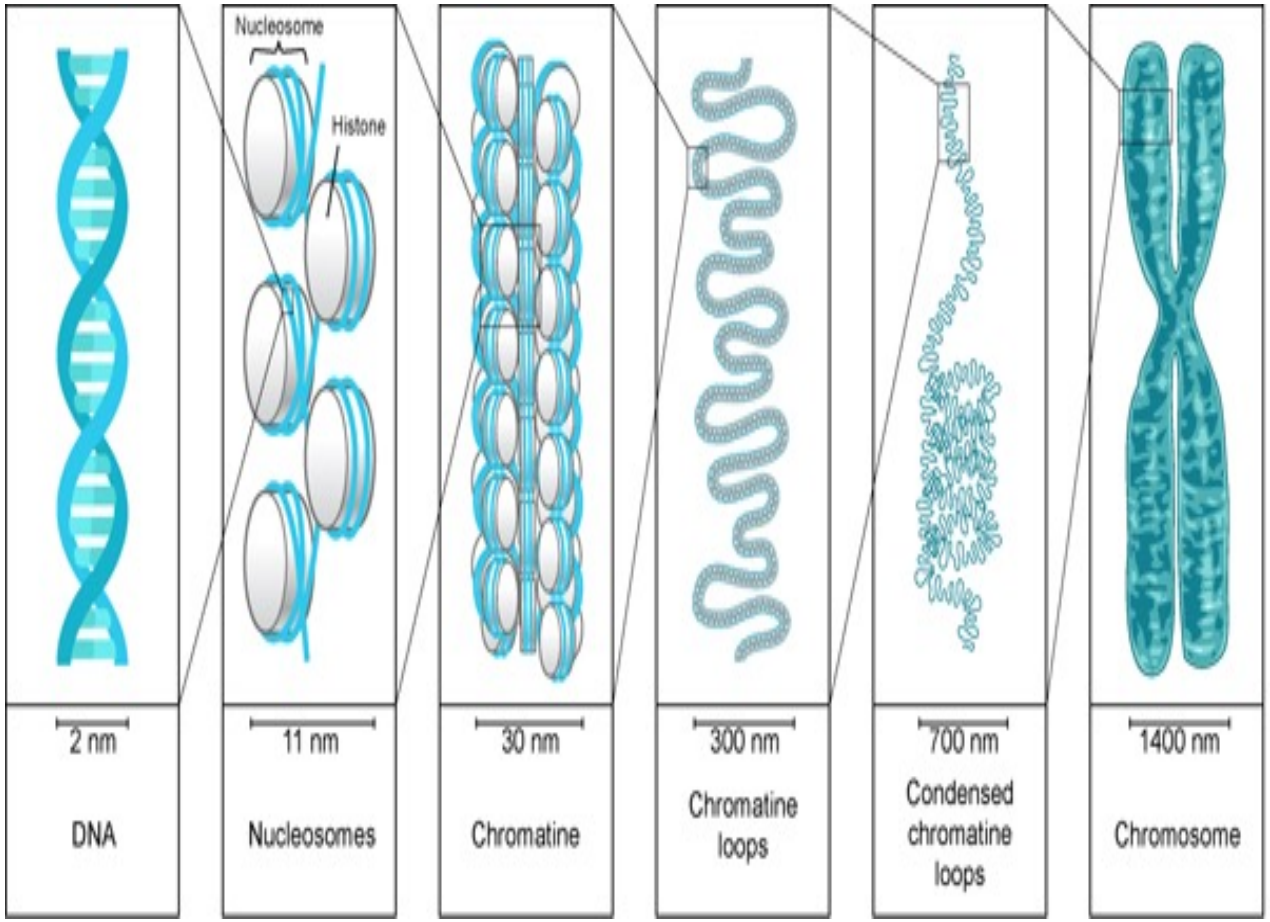
Higher-Order Structures: Nucleosomes fold into 30 nm chromatin fibre, further looped and scaffolded during mitosis.



Histone Type	Function	Location
H2A, H2B, H3, H4	Form the **histone octamer**	Core of the nucleosome
H1	**Linker histone**	Outside the core, on **linker DNA** and DNA entry/exit points

From Nucleosome to Chromosome: Chromatin Packing

- **Beads-on-a-String (11 nm fiber):**Linear array of nucleosomes (DNA wrapped around histone octamers).Histone H1 binds to the linker DNA and stabilizes the structure.
- **30 nm Fiber:** Nucleosomes fold into a more compact, helical fiber. This level of compaction is mediated by linker histone H1 and histone tail interactions.
- **Looped Domains (~300 nm):**The 30 nm fiber forms loops that attach to a chromatin scaffold. These loops help in spatial genome organization and gene regulation.
- **Metaphase Chromosome (~700 nm fiber per chromatid):**Looped domains undergo further folding during mitosis. Results in the highly condensed structure of visible chromosomes.

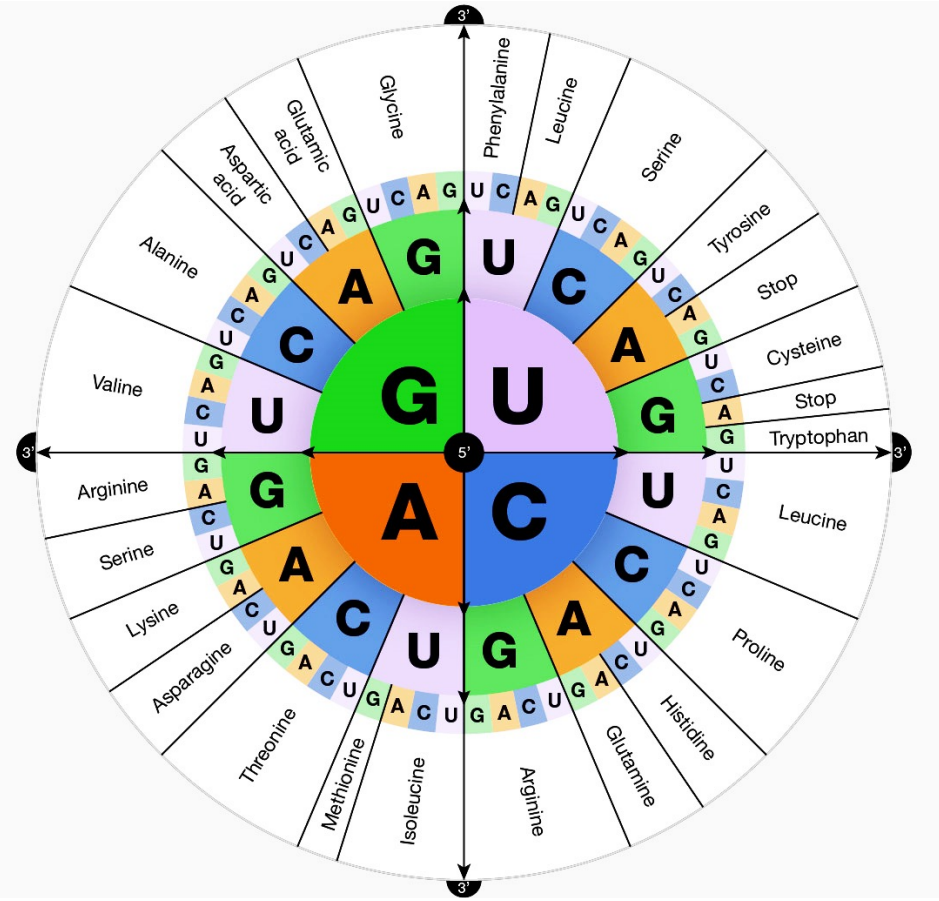


CONCEPT OF GENETIC CODE

Each gene's code uses the four nucleotide bases of DNA: **adenine (A), cytosine (C), guanine (G)** and **thymine (T)** — in various ways to spell out **three-letter "codons"** that specify which **amino acid** is needed at each position within a protein.

Properties of genetic code

- **Triplet Nature:** 64 possible codons from 4^3 combinations of nucleotides.
- **Degeneracy:** Multiple codons can encode the same amino acid (e.g., Leucine has 6 codons).
- **Non-overlapping:** Codons are read one after another without overlapping.
- **Unambiguous:** Each codon specifies only one amino acid.
- **Universal:** Consistent across most living organisms with few exceptions.



Start and Stop Codons in the Genetic Code

Start Codon:

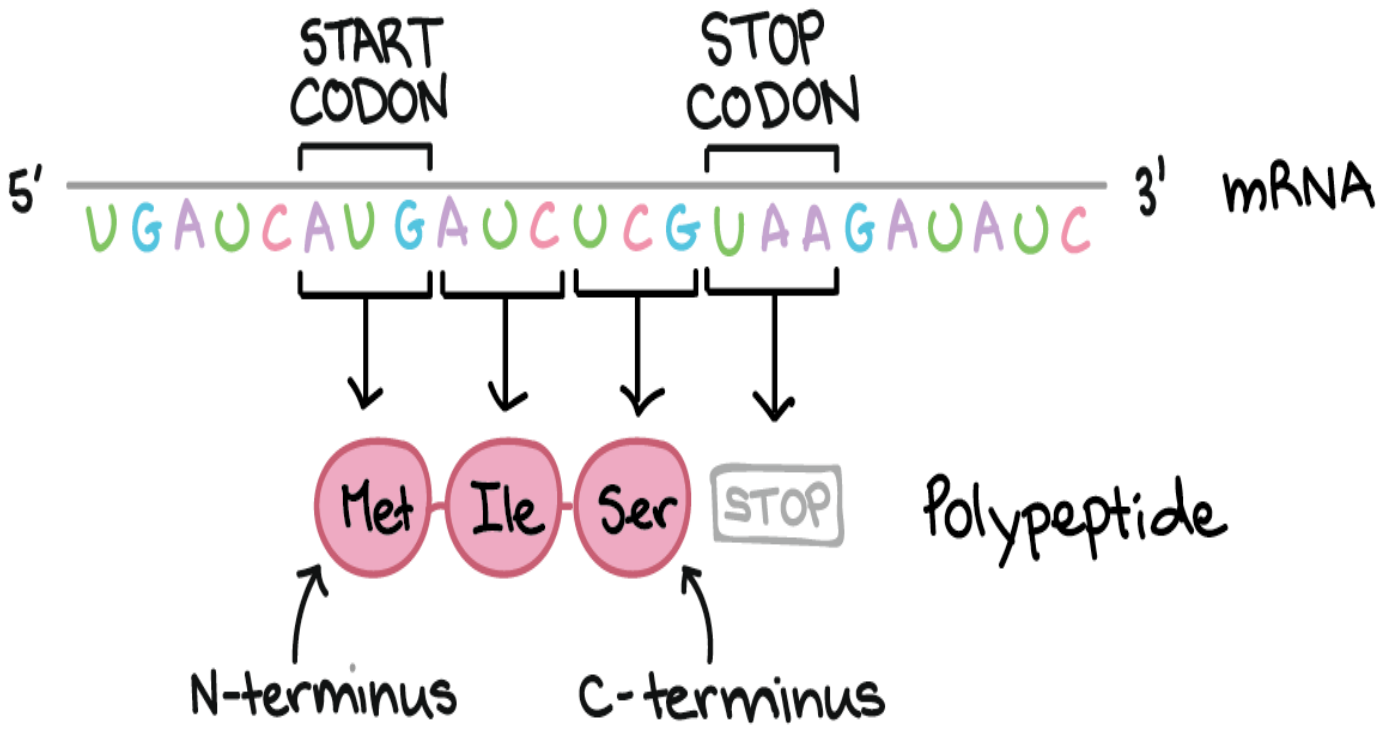
- **AUG** is the universal start codon.
- Codes for **methionine** in eukaryotes
- Signals the **initiation** of translation by ribosomes.

Stop Codons:

- **UAA, UAG, UGA** are stop codons.
- Do **not** code for any amino acid.
- Signal the **termination** of protein synthesis.
- Recognized by **release factors**, not tRNA.

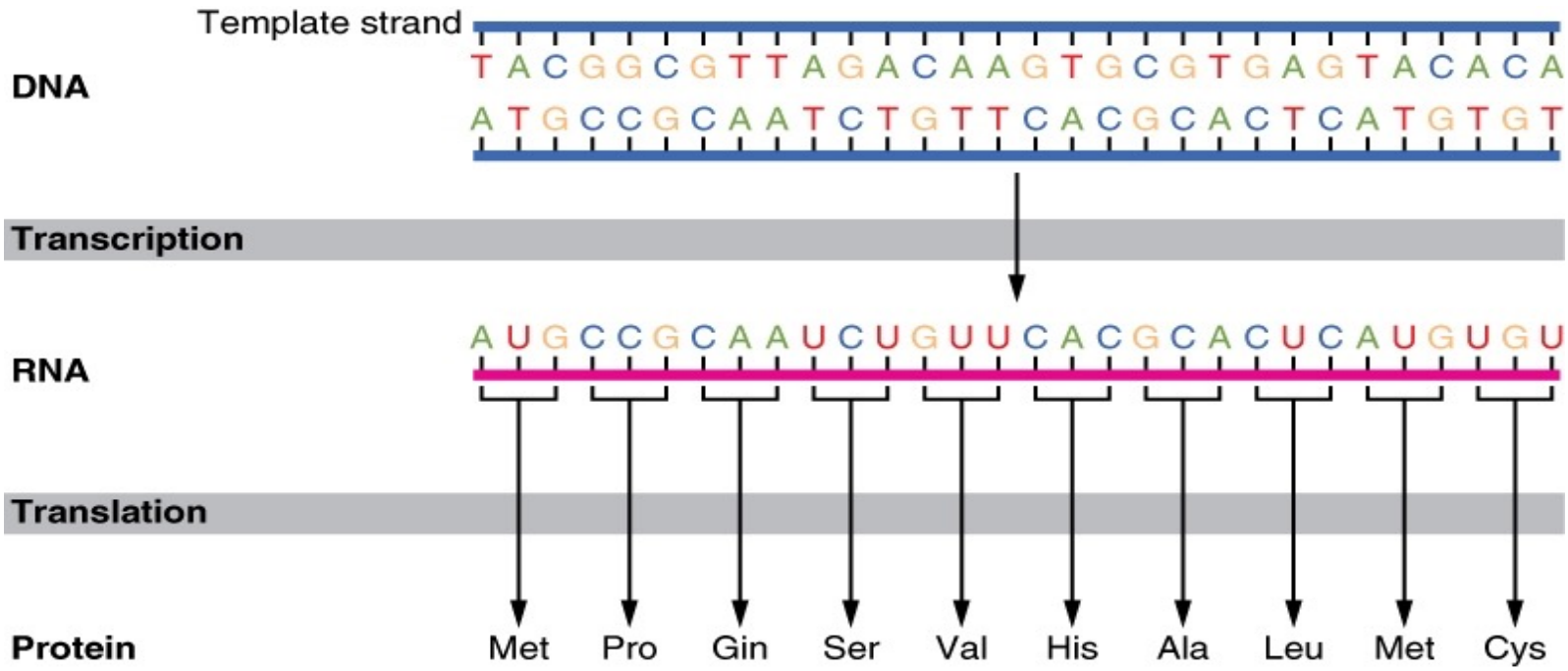
Biological Significance:

- Ensure proteins are synthesized **with correct boundaries**.
- Errors in recognition can lead to **truncated or extended proteins**, affecting function.



Genetic Code

- For **20 amino acids** there should be **20 codons**.
- Each codon should have **3 nucleotides** to impart specificity to each of the amino acid for a specific codon
- **1 Nucleotide:** 4 combinations
- **2 Nucleotides:** 16 combinations
- **3 Nucleotides:** 64 combinations (Most suited for 20 amino acids)
- **64 codons** in total and **3** out of these are **Non-Sense codons.**
- **61 codons** for **20 amino acids**



Functional Significance of the Genetic Code

- **Protein Synthesis:** The genetic code ensures that nucleotide sequences in genes are correctly translated into functional proteins.
- **Mutation Effects**: Changes in codons (mutations) can lead to altered amino acid sequences, potentially causing diseases (e.g., sickle cell anemia).
- **Biotechnological Applications:** Knowledge of the code enables recombinant protein production, gene editing (e.g., CRISPR), and synthetic biology.
- **Codon Optimization:** Different organisms prefer certain codons; optimizing codons enhances gene expression in heterologous systems.

TAT TGG CTA GTA CAT

Tyr Trp Leu Val His

TAT TGA CTA GTA CAT

Tyr STOP

Universality of the Genetic Code

- The genetic code is remarkably conserved across all domains of life—bacteria, archaea, and eukaryotes—which all use the same basic set of codon-to-amino acid assignments.
- This universality means that, for example, **AUG** codes for **methionine** and **UUU** codes for **phenylalanine** in almost all known organisms, from *Escherichia coli* to *Homo sapiens*.
- The widespread conservation strongly suggests a common evolutionary origin of life, where the genetic code was established in a universal ancestor and maintained throughout evolution.
- Despite this high degree of conservation, **minor variations** exist, often in specialized systems:

  - **In human mitochondria, the codon UGA, typically a stop codon in the standard code, encodes tryptophan (Trp).**
  - **AUA, which codes for isoleucine in the standard code, is read as methionine in human mitochondria.**

Organism/System	Codon	Standard Code	Reassigned To
Human mitochondria	UGA	Stop	Tryptophan (Trp)
	AUA	Isoleucine (Ile)	Methionine (Met)
	AGA, AGG	Arginine (Arg)	Stop

Degeneracy of Genetic code

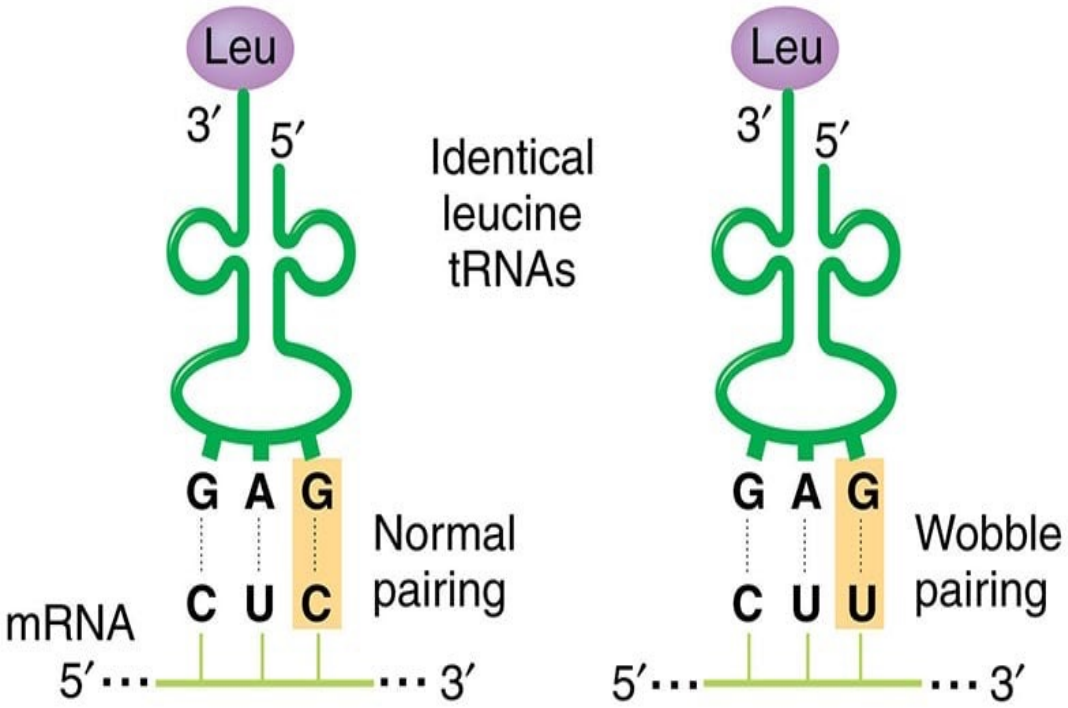
- **Degeneracy** refers to the **redundancy** of the genetic code, wherein **more than one codon** can specify the **same amino acid**.
- This feature provides **error tolerance** in the translation of genetic information.
- **Key Facts:**
- The genetic code consists of **64 codons** (4^4 combinations of A, U, G, and C).
- These codons encode **20 standard amino acids** and **3 stop signals**.
- As a result, **many amino acids are encoded by multiple codons**.
- **Examples of Degeneracy:**
  - **Leucine (Leu)**: encoded by **6 codons** (UUA, UUG, CUU, CUC, CUA, CUG)
  - **Serine (Ser)**: encoded by **6 codons**
  - **Arginine (Arg)**: encoded by **6 codons**
  - **Glycine (Gly)**: encoded by **4 codons**

Amino Acid			Codons					
W	Trp	Tryptophan	TGG					
Y	Tyr	Tyrosine	TAC	TAT				
C	Cys	Cysteine	TGC	TGT				
E	Glu	Glutamic acid	GAA	GAG				
K	Lys	Lysine	AAA	AAG				
Q	Gln	Glutamine	CAA	CAG				
S	Ser	Serine	AGC	AGT	TCA	TCC	TCG	TCT
L	Leu	Leucine	TTA	TTG	CTA	CTC	CTG	CTT
R	Arg	Arginine	AGA	AGG	CGA	CGC	CGG	CGT
G	Gly	Glycine	GGA	GGC	GGG	GGT		
F	Phe	Phenylalanine	TTC	TTT				
D	Asp	Aspartic acid	GAC	GAT				
H	His	Histidine	CAC	CAT				
N	Asn	Asparagine	AAC	AAT				
M	Met	Methionine	ATG					
A	Ala	Alanine	GCA	GCC	GCG	GCT		
P	Pro	Proline	CCA	CCC	CCG	CCT		
T	Thr	Threonine	ACA	ACC	ACG	ACT		
V	Val	Valine	GTA	GTC	GTG	GTT		
I	Ile	Isoleucine	ATA	ATC	ATT			
X	STP	Stop codon	TAA	TAG	TGA			

Wobble Hypothesis

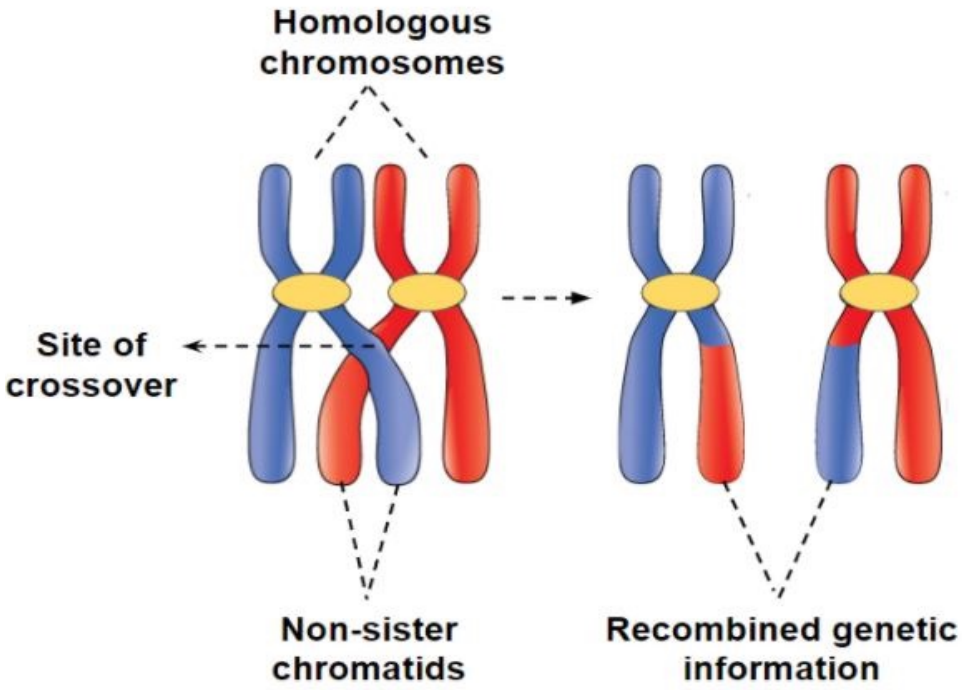
- The Wobble Hypothesis explains how a **limited number of tRNA** molecules can recognize **multiple codons** for **the same amino acid** during protein synthesis.
- It states that the **third base** of a codon on mRNA **can pair with more than one base at the first position** of the tRNA anticodon. This "wobble" pairing, which is **less strict** than the standard Watson-Crick base pairing, **allows for the degeneracy of the genetic code.**

- **This contributes to:**
- **Minimizing the effect of point mutations**—a mutation in the third base may **still result in the same amino acid** (silent mutation).
- **Stability and efficiency** in protein synthesis.
- **Evolutionary flexibility** without altering protein function.

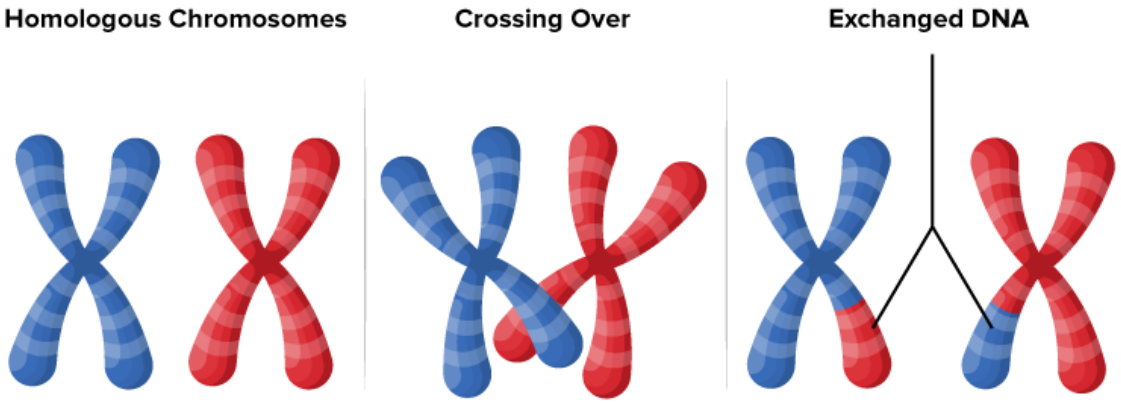


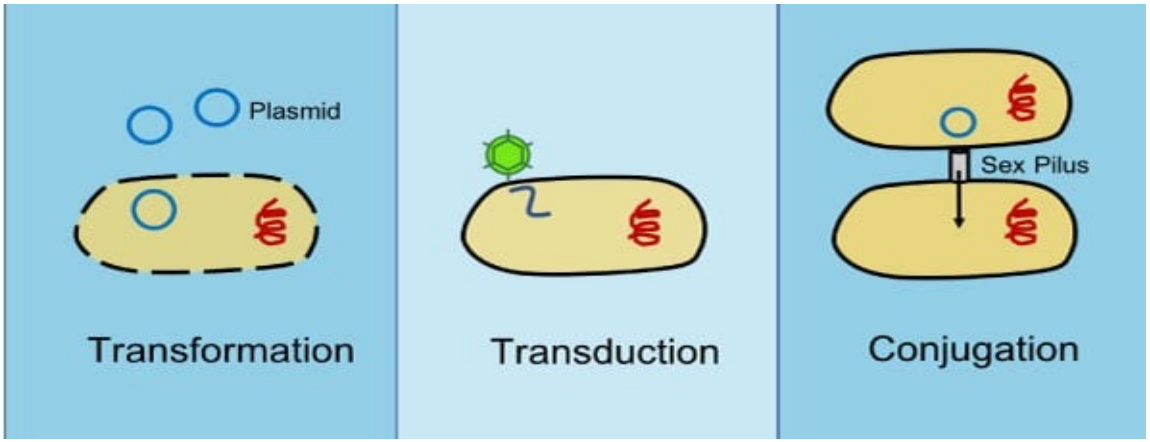
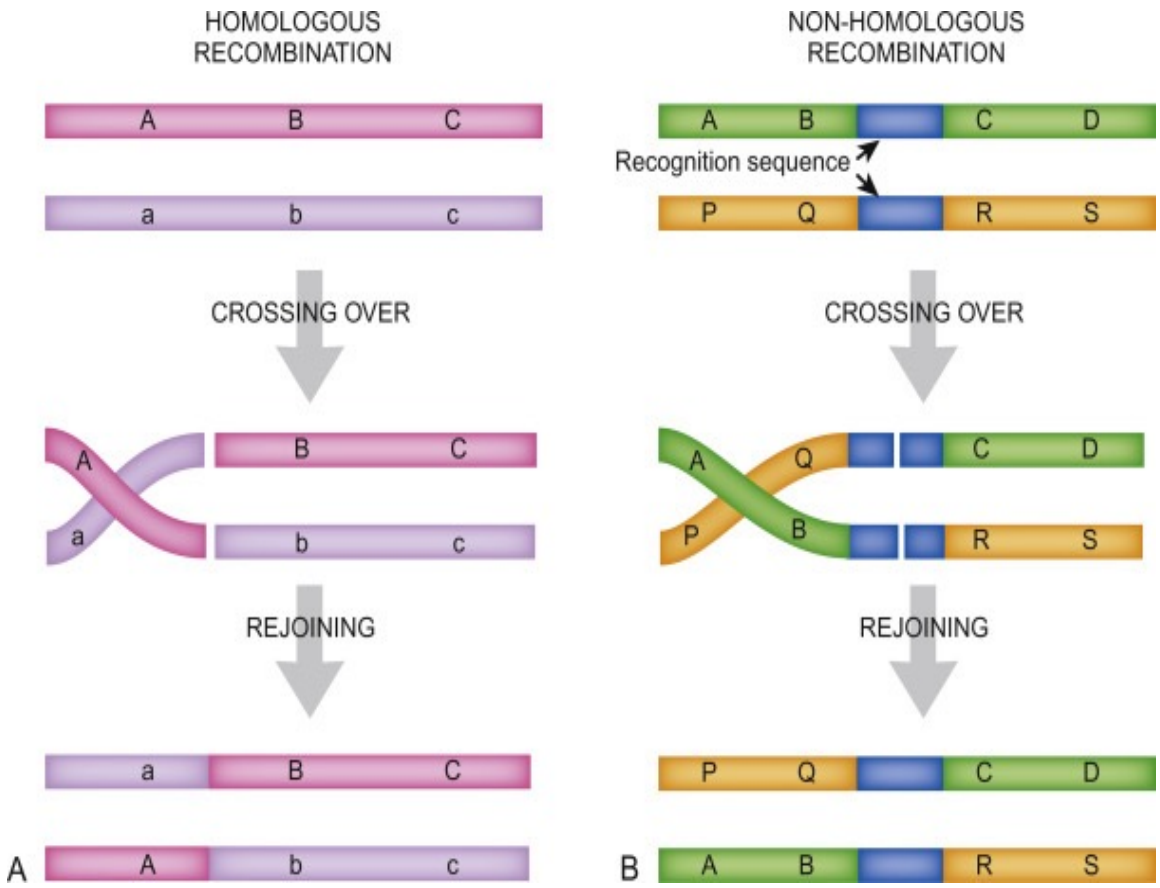
Gene Definition in Terms of Recombination

- Genetic recombination refers to the process where organisms **exchange genetic material**, resulting in offspring with **unique combinations** of **traits not found in either parent**. This exchange can occur between different regions within the same chromosome or between different chromosomes.
- Recombination involves the shuffling and mixing of genetic information. This can happen during **sexual reproduction (meiosis)** or through other mechanisms like **transformation, transduction,** and **conjugation.**
- Recombination is a **major source of genetic variation** within populations, **allowing for adaptation to changing environments.**



- **Examples:**
  - **Meiosis (Crossing Over):** During meiosis, homologous chromosomes (paired chromosomes from the mother and father) can exchange segments of DNA through a process called crossing over, according to Nature. This results in new combinations of alleles (different versions of genes) on the chromosomes.
  - **Bacteria:** Bacteria can also exchange genetic material through mechanisms like transformation (taking up DNA from the environment), transduction (via bacteriophages), and conjugation (direct transfer between bacteria).





- **Types of Recombination:**
  - **Homologous Recombination:** This is a common type of recombination where similar DNA sequences exchange information.
  - **Non-homologous Recombination:** This type of recombination can occur between DNA sequences that are not closely related.
- **Significance:**
  - **Evolutionary Adaptation:** Recombination provides raw material for natural selection to act upon, leading to the evolution of new traits and increased adaptability.
  - **Genetic Diversity:** Recombination is crucial for maintaining genetic diversity within populations, making them more resilient to environmental changes.
  - **Repair of DNA Damage:** Recombination can also play a role in repairing damaged DNA.